ELITERS

Consolidated Document

Problem Number: 17

Problem Statement:
Choosing the Next Expansion City

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1. Problem understanding

a. Real-World Context

Expanding into a new city is one of the most critical growth decisions for any brand. It involves committing significant resources capital investment, supply chain setup, hiring, marketing, and operations often with limited certainty about outcomes.

In today's competitive and data-driven environment, cities differ widely in:

- Consumer demand and purchasing power
- Population growth and urbanization trends
- Infrastructure readiness (transport, logistics, digital access)
- Competitive intensity and market saturation

Despite the availability of data, the decision is often fragmented and subjective**,** with different teams prioritizing different factors. For example:

- Marketing may prioritize demand and demographics
- Finance may focus on cost and ROI
- Operations may emphasize logistics and feasibility

This misalignment leads to decision, internal conflicts, and increased risk of poor expansion choices**.**

Target Users / Stakeholders

- **Business Leaders / Strategy Teams:**
  Responsible for making the final expansion decision and aligning it with long-term growth strategy
- **Investors / Financial Stakeholders:**
  Interested in maximizing returns while minimizing risk exposure
- **Operations Teams:**
  Focused on execution feasibility, supply chain efficiency, and infrastructure readiness
- **Marketing & Sales Teams:**
  Concerned with customer demand, brand fit, and competitive positioning
- **End Customers:**
  Indirect stakeholders who benefit from improved accessibility, better services, and localized offerings

Why This Problem Matters

- A wrong expansion decision can lead to financial losses, brand dilution, and operational inefficiencies
- A well-informed decision can unlock new revenue streams, market leadership, and long-term scalability
- Increasing data availability makes it possible—but not easy—to make evidence-based decisions instead of intuition-driven ones

2. Success Criteria

Success is achieved when the chosen city demonstrates strong early performance, sustainable growth potential, and controlled risk, while the decision itself is transparent, data-backed, and aligned across all stakeholders**.**

A successful solution should enable the brand to make a data-driven, low-risk, and high-impact expansion decision**.** The effectiveness of the solution can be evaluated using the following measurable outcomes:

1. Financial Performance

- Projected ROI ≥ Target Benchmark (e.g., ≥ 20–25% within first 2–3 years)
- Break-even Period ≤ Defined Timeline (e.g., within 12–24 months)
- Revenue Growth Rate aligned with expansion goals

2. Market Potential & Demand Fit

- High market demand score based on population size, income levels, and category demand
- Strong customer acquisition rate in the initial launch phase
- Evidence of product-market fit in the selected city

3. Competitive Advantage

- Entry into markets with manageable competition levels
- Clear differentiation opportunities compared to existing competitors
- Ability to capture market share within defined timeframe

4. Operational Feasibility

- Availability of adequate infrastructure (logistics, transportation, supply chain)
- Operational cost within budget constraints
- Smooth setup with minimal delays in execution

5. Risk Minimization

- Identification and mitigation of key risks (regulatory, market, operational)
- Selection of cities with stable economic and regulatory environments
- Reduced likelihood of market failure or exit scenarios

6. Decision Quality & Alignment

- Transparent and justifiable decision-making process
- Alignment across stakeholders (strategy, finance, operations)
- Ability to replicate the decision model for future expansions

3. Prompt set (3–5 prompts total)

The following prompts (P1-P5) are designed to address the problem systematically. Each prompt serves a specific purpose in the workflow.

P1: Problem Structuring & Variable Identification

Purpose: Transforms the ambiguous business problem into a structured framework by identifying key variables and organizing them into decision categories.

Prompt:
You are a senior Business Intelligence Consultant.

Analyze the problem of selecting the best city for expansion and convert it into a structured decision framework.

Tasks:

1. Identify all relevant decision variables (demographics, income levels, demand patterns, infrastructure, competition, costs, regulatory environment, etc.)
2. Group these variables into logical categories such as:
   - Market Potential
   - Financial Viability
   - Operational Feasibility
   - Risk Factors
3. Define measurable indicators or proxies for each variable
4. Highlight dependencies and trade-offs between variables
5. Present the output in a clear, structured format suitable for executive decision-making

Ensure the framework is comprehensive, realistic, and applicable to real-world business scenarios.

<u>P2:</u> Weighted Scoring & Decision Model

Purpose: Converts the structured framework into a quantitative model to objectively compare multiple cities.

Prompt:
You are a data-driven strategy analyst.

Using the structured framework provided, design a weighted scoring model to evaluate and compare candidate cities.

Tasks:

1. Assign weights to each category based on strategic importance
2. Define a scoring scale (e.g., 1–10) for each metric
3. Create a formula to calculate a final composite score for each city
4. Explain how trade-offs are handled (e.g., high demand vs high competition)
5. Provide a sample scoring table format for at least 3 cities

Ensure the model is transparent, logical, and adaptable to different business priorities.

<u>P3:</u> Multi-Expert Evaluation & Risk Analysis

Purpose: Introduces multiple perspectives and identifies risks to reduce blind spots in decision-making.

Prompt:
You are a panel of experts consisting of:

- Market Analyst
- Financial Analyst
- Operations Manager
- Risk Consultant

Evaluate the city selection decision from each expert's perspective.

Tasks:

1. Each expert identifies their top 3 evaluation priorities
2. Analyze potential risks such as:
    - Market saturation
    - Demand overestimation
    - Infrastructure limitations
    - Regulatory challenges

3. Highlight disagreements or conflicting priorities among experts
4. Provide:
    - Best-case scenario
    - Worst-case scenario
    - Most likely scenario
5. Deliver a balanced recommendation considering all viewpoints

Ensure the analysis reflects real-world uncertainty and trade-offs.

P4: Critical Review & Model Refinement

Purpose: Challenges assumptions, identifies weaknesses, and improves the robustness of the decision model

Prompt:
You are a critical reviewer of business decision frameworks.

Evaluate the proposed city selection model and identify its limitations.

Tasks:

1. Identify assumptions that may be weak, biased, or unsupported
2. Detect any bias in weighting, scoring, or data interpretation
3. Highlight missing variables or overlooked risks
4. Suggest improvements to enhance accuracy and reliability
5. Provide a refined version of the decision-making approach

Be direct, analytical, and constructive. Focus on improving decision quality rather than validating the existing model.

P5: Final Recommendation & Executive Summary

Purpose: Synthesizes all prior analysis into a clear, actionable recommendation for decision-makers.

Prompt:

You are a senior strategy consultant presenting to executives.

Based on the structured framework, scoring model, and expert analysis, provide a final recommendation for the best city to expand into.

Tasks:

1. Identify the top-ranked city and justify the selection
2. Summarize key strengths and risks of the chosen city

3. Compare it briefly with alternative options
4. Provide actionable next steps for implementation
5. Present the output as a concise executive summary

Ensure the recommendation is clear, confident, and business-focused.

4. Prompt evolution notes

This section explains how the prompts relate to each other and how they improve the workflow:

1. P1 to P2

P1 converts the vague business problem into a clear and structured framework by identifying key variables and organizing them into categories.

P2 builds directly on this by:

- Assigning weights and scores to the variables defined in P1
- Converting qualitative insights into a quantitative decision model
- Enabling objective comparison across multiple cities

Improvement: P1 provides clarity; P2 adds measurability and decision-making capability**.**

2. P2 to P3

P2 produces a structured and data-driven scoring system, but it has limitations:

- It assumes all variables and weights are correctly defined
- It may overlook real-world uncertainties and conflicting priorities

P3 addresses these gaps by:

- Introducing **multiple expert perspectives** (market, finance, operations, risk)
- Identifying **hidden risks and edge cases**
- Highlighting **conflicts and trade-offs** that a scoring model may ignore
- Adding **scenario analysis** (best, worst, most likely)

Improvement:
P2 ensures objectivity; P3 adds **depth, realism, and risk awareness**.

3. Overall Workflow

The prompts work together as a **layered decision system**, moving from ambiguity to a robust final recommendation:

1. **P1 – Structuring:**
   Defines the problem clearly and identifies all relevant variables
2. **P2 – Quantification:**
   Converts the framework into a **scoring model for comparison**
3. **P3 – Multi-Perspective Analysis:**
   Adds **expert insights, risk evaluation, and scenario planning**
4. **P4 – Critique & Refinement:**
   Challenges assumptions and improves the model's reliability
5. **P5 – Final Recommendation (Optional):**
   Synthesizes all insights into a **clear, actionable decision**

5. Ethical and responsible AI considerations

a. Main Risks in This Problem Domain

One major risk is **data bias**, where available data may be incomplete, outdated, or skewed toward certain regions or income groups. This can lead to incorrect assumptions about market potential and customer demand.

Another risk is **misleading assumptions**, such as assuming past growth trends will continue in the future. Real-world factors like economic shifts, policy changes, or unexpected disruptions can make these assumptions unreliable.

There is also a risk of **oversimplification**, where complex decisions are reduced to scoring models. This can ignore qualitative factors like cultural fit, local behavior, or brand perception, which are critical for success.

Additionally, **overconfidence in AI outputs** can be dangerous. Decision-makers might treat model results as final answers instead of using them as guidance, leading to poor strategic choices.

Finally, **hidden bias in weighting and data gaps** can affect outcomes. If weights are assigned subjectively or important data is missing, the decision model may produce misleading results.

b. How your prompts or workflow reduce these risks.

To reduce these risks, the workflow begins with **clear variable definition (P1)**, ensuring all important factors are explicitly identified. This makes assumptions visible and easier to challenge.

The **scoring model (P2)** is designed to be transparent and adjustable, allowing stakeholders to understand and modify weights based on business priorities. This reduces hidden bias.

The inclusion of **multiple expert perspectives (P3)** helps bring diverse viewpoints into the analysis. It highlights risks, conflicts, and blind spots that a single model might miss.

Scenario analysis (P3) further strengthens the approach by considering best-case, worst-case, and most-likely outcomes, preventing over-reliance on a single prediction.

Finally, the **critique step (P4)** ensures continuous improvement by identifying weaknesses, biases, and missing factors. Combined with human oversight, this makes the system more reliable and responsible.

6. Short conclusion

a. Strengths of Our Prompt Strategy

This prompt strategy stands out because it converts a vague business problem into a **clear, structured, and repeatable decision-making process**. Instead of jumping to conclusions, it builds the solution step-by-step—starting from problem structuring, moving to quantification, and then adding expert perspectives and critique.

Another key strength is its **balance between qualitative and quantitative analysis**. It does not rely only on numbers but also incorporates real-world thinking through multi-expert evaluation and scenario analysis.

The approach also emphasizes **transparency and explainability**. Every assumption, weight, and decision factor is visible, making it easier for stakeholders to understand and trust the outcome.

Finally, the inclusion of a critique and refinement stage ensures that the model is not blindly accepted. This makes the overall strategy more robust, realistic, and aligned with real business decision-making.

b. Limitations and Trade-offs

Despite its strengths, the strategy still depends heavily on the quality and availability of input data. If the data is inaccurate or incomplete, the outputs will also be unreliable.

The use of scoring models introduces simplification of complex real-world dynamics. Not all factors can be perfectly quantified, and some important qualitative insights may still be overlooked.

There is also a subjectivity trade-off in assigning weights and priorities. Even with transparency, different stakeholders may disagree on what matters most, which can influence the final outcome.

Additionally, the process can be time-intensive, especially when involving multiple perspectives and iterative refinement. This may not always be practical in fast-paced decision environments.